SAFETY DATA SHEET

Revision Date: 00/08-06-2016

Conforms to UN GHS Classification

SECTION 1. IDENTIFICATION

Trade name : **BACTERCID B20**

Product type : Liquid

Material uses : Other non-specified industry: In-can preservative.

Manufacturer : NP-Tech Company Limited
14-231, Hanyang University, 17 Haengdang-dong,
Seongdong-gu, Seoul, 133-791, Korea.
Tel : 82-2-2220-0498
Fax : 82-2-2291-0498

SECTION 2. HAZARDS IDENTIFICATION

Classification of the substance or mixture : ACUTE TOXICITY (oral) – Category 5
SKIN CORROSION/IRRITATION – Category 1
SERIOUS EYE DAMAGE/EYE IRRITATION – Category 1
SKIN SENSITIZATION – Category 1
AQUATIC HAZARD (ACUTE) – Category 2

GHS label elements

Hazard pictograms :



Signal word : Danger

Hazard statements : May be harmful if swallowed.
Causes severe skin burns and eye damage.
May cause an allergic skin reaction.
Toxic to aquatic life.

Precautionary statements

Prevention : Wear protective gloves. Wear eye or face protection. Wear protective clothing. Avoid release to the environment. Avoid breathing vapor. Wash hands thoroughly after handing. Contaminated work clothing should not be allowed out of the workplace.

Response : IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER or physician.
IF SWALLOWED: Immediately call a POISION CENTER or physician. Rinse mouth. DO NOT induce vomiting.
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. Wash contaminated clothing before reuse. Immediately call a POISION CENTER or physician.
IF ON SKIN: Wash with plenty of soap and water. Take off contaminated clothing and wash it before reuse. If skin irritation or rash occurs: Get medical attention.
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISION CENTER or physician.

Storage : Store locked up.

Disposal : Dispose of contents and container in accordance with all local, regional, national and international regulations.

Other hazards which do not result in classification : None known.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance/mixture : Mixture
Other means of Identification : Not available.

CAS number/other identifiers
CAS number : Not applicable.
EC number : Mixture

Ingredients	CAS number	%
1,2 Benzisothiazolin-3-one	2634-33-5	15-20
Glycerin	56-81-5	25-50
1,2 Propane diol	57-55-6	25-50
Sodium hydroxide	1310-73-2	5-7
Water	7732-18-5	15-20

SECTION 4. FIRST AID MEASURES

Description of necessary first aid measures

Eye contact : Get medical attention immediately. Immediately flush eyes with plenty of water, occasionally lifting the upper and lower eyelids. Check for and remove any contact lenses. Continue to rinse for at least 10 minutes. Chemical burns must be treated promptly by a physician.

Inhalation : Get medical attention immediately. Move exposed person to fresh air. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. Keep person warm and at rest. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained

personnel. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband.

Skin contact : Get medical attention immediately. Flush contaminated skin with plenty of water. Continue to rinse for at least 10 minutes. Remove contaminated clothing and shoes. Wash contaminated clothing thoroughly with water before removing or wear gloves. Chemical burns must be treated promptly by a physician. Wash clothing before reuse. Clean shoes thoroughly before reuse.

Ingestion : Get medical attention immediately. Wash out mouth with water. Remove dentures if any. Move exposed person to fresh air. Keep person warm and at rest. If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Stop if the exposed person feels sick as vomiting may be dangerous. Do not induce vomiting unless directed to do so by medical personnel. If vomiting occurs, the head should be kept low so that vomit does not enter the lungs. Chemical burns must be treated promptly by a physician. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband.

SECTION 5. FIRE-FIGHTING MEASURES

Hazardous thermal Decomposition products : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
nitrogen oxides
sulfur oxides
metal oxide/oxides

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions : Immediately contact emergency personnel. Keep unnecessary personnel away. Use suitable protective equipment.

Environmental precautions : Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers.

Methods for cleaning up : If emergency personnel are unavailable, contain spilt material. For small spills, add absorbent (soil may be used in the absence of other suitable materials), scoop up material and place in a sealable, liquid-proof container for disposal. For large spills, dyke spilt material or otherwise contain material to ensure runoff does not reach a waterway. Place spilt material in an appropriate container for disposal.

Note: see section 8 for personal protective equipment and section 13 for waste disposal.

SECTION 7. HANDLING AND STORAGE

Handling : Do not get in eyes or on skin or clothing. Keep container closed. Use only with adequate ventilation. Do not breathe vapor or mist. Avoid contact of spilt material and runoff with soil and surface waterways. Wash thoroughly after handling.

Storage : Keep container tightly closed. Keep container in a cool, well-ventilated area.

Packaging materials

Recommended use : Use original container.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Control parameters

Occupational exposure limits

Ingredient name	Exposure limits
Sodium hydroxide	**ACCGIH TLV (United States, 1/2009).** C: 2 mg/m^3

Appropriate engineering controls : If user operations generate dust, fumes, gas, vapor or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits.

Environmental exposure : Emission from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

Personal protective equipment

Respiratory system : Use a properly fitted, air-purifying or air-fed respirator complying with an approved standard if a risk assessment indicates this is necessary. Respirator selection must be based on known or anticipated exposure levels, the hazards of the product and the safe working limits of the selected respirator.

Skin and body : Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.

Hands : Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary. EU Standard for gloves: EN374

Eyes : Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists or dusts. Recommended: splash goggles, face shield

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	:	Transparent liquid
Color	:	Light yellow to light brown
Odor	:	Characteristic
pH	:	11.5-13.0 (as is)
Viscosity @ 25°C	:	< 500 cps.
Boiling point	:	100°C (212°F)
Melting point	:	Not available
Flash point	:	Not available
Explosive properties	:	Not available
Oxidizing properties	:	Not available
Vapor density (Air = 1)	:	Not available

Solubility in water	:	Soluble
Evaporation rate (Butyl acetate = 1)	:	< 1
% Volatile (by volume)	:	74
Specific Gravity @ 25°C	:	1.10-1.13

SECTION 10. STABILITY AND REACTIVITY

Reactivity	:	No specific test data related to reactivity available for this product or its ingredients.
Chemical stability	:	The product is stable.
Possibility of hazardous Reactions	:	Under normal conditions of storage and use, hazardous reactions will not occur.
Conditions to avoid	:	No specific data.
Incompatible materials	:	Reactive or incompatible with the following materials: acids
Hazardous decomposition products	:	Under normal conditions of storage and use, hazardous decomposition products should not be produced.

SECTION 11. TOXICOLOGICAL INFORMATION

Information on toxicological effects

Acute toxicity

Product/ingredient name	Result	Species	Dose	Exposure
Benzisothiazol-3(2H)-one	LD50 Dermal	Rat	> 2000 mg/kg	-
	LD50 Oral	Mouse	1150 mg/kg	-
	LD50 Oral	Rat	597 mg/kg	-

Product/ingredient name	Result	Species	Dose	Exposure
Sodium hydroxide	LD50 Dermal	Rabbit	1350 mg/kg	-
	LD50 Oral	Rat	2000 mg/kg	-
	LD50 Oral	Rabbit	500 mg/kg	-

Irritation/Corrosion

Product/ingredient name	Result	Species	Score	Exposure	Observation
Benzisothiazol-3(2H)-one	Eyes - Severe irritant	Rat	-	-	-
	Skin – Moderate irritant	Rabbit	-	-	-

Sensitization

Product/ingredient name	Route of exposure	Species	Result
Benzisothiazol-3(2H)-one	Skin	Rabbit	Sensitizing

Mutagenicity : Not available
Carcinogenicity : Not available
Reproductive toxicity : Not available
Teratogenicity : Not available
Specific target organ toxicity (single exposure) : Not available
Specific target organ toxicity (repeated exposure) : Not available
Aspiration hazard : Not available
Information on the likely routes of exposure : Not available

Potential acute health effects

Eye contact : Causes serious eye damage.
Inhalation : No known significant effect or critical hazards.
Skin contact : Causes severe burns. May cause an allergic skin reaction.
Ingestion : May be harmful if swallowed.

Symptoms related to the physical, chemical and toxicological characteristics.

Eye contact	:	Adverse symptoms may include the following: pain watering redness
Inhalation	:	No specific data.
Skin contact	:	Adverse symptoms may include the following: pain or irritation redness blistering may occur
Ingestion	:	Adverse symptoms may include the following: Stomach pains

Delayed and immediate effects and also chronic effects from short and long term exposure

Short term exposure

Potential immediate effects	:	Not available
Potential delayed effects	:	Not available

Long term exposure

Potential immediate effects	:	Not available
Potential delayed effects	:	Not available

Potential chronic health effects

General	:	Once sensitized, a severe allergic reaction may occur when subsequently exposed to very low levels.
Carcinogenicity	:	No know signification effects or critical hazards.
Mutagenicity	:	No know signification effects or critical hazards.
Teratogenicity	:	No know signification effects or critical hazards.
Developmental effects	:	No know signification effects or critical hazards.
Fertility effects	:	No know signification effects or critical hazards.

SECTION 12. ECOLOGICAL INFORMATION

Toxicity

Product/ingredient name	Result	Species	Exposure
Benzisothiazol-3(2H)-one	Acute EC50 2.44 mg/l	Daphnia – Daphnia magna	48 hours
	Acute LC50 0.74 mg/l	Fish	96 hours
	Acute EC50 76 mg/l	Daphnia	24 hours
	Acute EC50 40.38 mg/l Fresh water	Daphnia – Ceriodahnia dubia – Neonate	48 hours
	Acute LC50 33000 to 100000 µg/l Marine water	Crustaceans – Crangon crangon – Adult	48 hours
	Acute LC50 99 mg/l	Fish – Lepomis macrochirus	48 hours
	Acute LC50 45.5 mg/l	Fish – Oncorhynchus mykiss	96 hours
	Chronic NOEC 56 mg/l Marine water	Fish – Poecilia reticulate - Young	96 hours

Persistence and degradability : Not available

Bioaccumulative potential

Product/ingredient name	$LogP_{ow}$	BCF	Potential
Benzisothiazol-3(2H)-one	0.4	-	low

Mobility in soil

Soil/water partition Coefficient (K_α) : Not available

Other adverse effects : No known significant effects or critical hazards.

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods : The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

SECTION 14. TRANSPORT INFORMATION

	UN	IMDG	IATA
UN number	UN3266	UN3266	UN3266
UN proper shipping name	Corrosive liquid, basic, inorganic. n.o.s. (Contains: Sodium hydroxide)	Corrosive liquid, basic, inorganic. n.o.s. (Contains: Sodium hydroxide)	Corrosive liquid, basic, inorganic. n.o.s. (Contains: Sodium hydroxide)
Transport hazard class(es)	8 	8 	8 

	UN	IMDG	IATA
Packing group	III	III	III
Environmental hazards	No.	No.	No.
Additional information	-	**Emergency schedules (EmS)** F-A, S-B	The environmentally hazardous substance mark may appear if required by other transportation regulations.

Special precaution for user : **Transport within user's premises:** always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

Transport in bulk according To Annex II of MARPOL 73/78 and the IBC Code : Not available

SECTION 15. REGULATORY INFORMATION

Safety, health and environmental regulations specific for the product : No known specific national and/or regional regulations applicable to this product (including its ingredients).

Federal registrations

Toxic substances control act (TSCA inventory) status : Not found on the U.S. EPA TSCA inventory. For use only in products that are not regulated under TSCA (such as EPA FIFRA registered and EPA regulated products), or to be used in research and development 9R&D) only by technically qualified individuals.

TSCA (Toxic Substances Control Act) Regulations, 40 CFR 710	:	All ingredients are on the TSCA Chemical Substance Inventory.
CERCLA (Comprehensive Environmental Response, Compensation and Liability Act of 1980) and SARA Regulations (40CFR 355, 370, and 372)	:	This product does not contain any chemicals subject to the reporting requirements of SARA Section 313.
Canadian Domestic Substance List	:	This product is contained on the Canadian Domestic Substance List.

SECTION 16. OTHER INFORMATION

History

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